

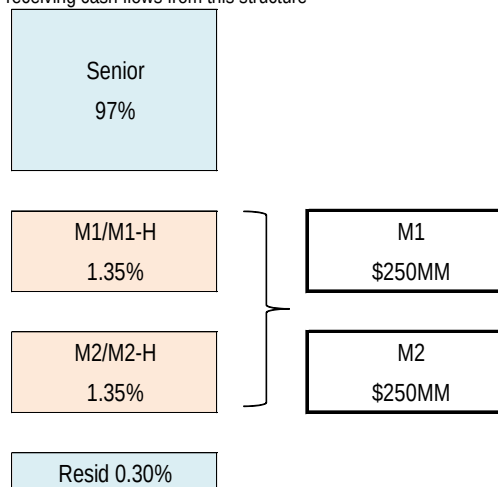
John Sim (1-212) 834-3124
Abhishek Mistry^{AC} (1-212) 834-4662

GSE Risk-Sharing: Mortgage credit exposure with agency collateral

- **Fannie Mae and Freddie Mac have issued five risk-sharing deals to date. The deals provide mortgage credit exposure to the GSE's wrap business**
- **Bonds are unsecured obligations of the GSEs with a 10-year final maturity, but the cash flows are linked to payments of a reference pool of mortgages**
- **Principal is paid pro-rata between the senior and subordinate bonds but sequentially within the subs**
- **Lockout can occur if various triggers are hit. Both deals accelerate principal to seniors if subordination falls below a target, but the Freddie deals contain an additional trigger based on credit events that can cause extension**
- **Defaults are replaced with credit events, which are generally given by 180-day delinquency. A fixed schedule based on cumulative credit events defines severities**
- **Deals so far have focused on very high credit borrowers, with LTVs below 80, no mortgage insurance, and no HARP loans. The GSEs are likely to expand the credit box in future deals**
- **We simplified our non-agency loan-level transition model and fit to Freddie delinquency data. Our new GSE credit model sheds a light on defaults across the new structures**
- **In addition to credit risk, prepayment risk affects cash flows, especially since the earlier bonds are trading at high premiums. Using our pool-level prepayment model, the prepay risk looks similar to TBA 3.5s**
- **We compare the GSE credit deals to other structured credit, including CMBS, CLOs, and 2.0 non-agencies. Looking at spread versus average life provides a convenient framework**

Exhibit 1: STACR 2013-DN1 - Freddie created two mezzanine risk classes from reference pools of collateral

The left side are not actual bonds. Only the M1 and M2 are certificates receiving cash flows from this structure



Source: J.P. Morgan, deal documents

As part of FHFA's initiative to bring private capital back to the mortgage market, Fannie and Freddie have issued several risk-sharing deals, where part of the credit risk on the GSE's wrap business is transferred to private investors in these deals. Freddie Mac debuted its first risk-sharing security (STACR 2013-DN1) in July 2013. Fannie Mae followed up with their version (CAS 2013-C01) in October. There have been five such deals issued by both GSEs to date. This new asset class has received wide attention by investors as this new class of bond presented opportunities for investors to access US residential mortgage credit, especially given weak issuance in 2.0 non-agencies. The GSEs are each planning to issue at least one deal per quarter going forward; in all, we expect around \$8bn of issuance in 2014.

Deal structure

The securities are floaters that are unsecured obligations of the GSEs with a 10-year maturity. Although cash flows are linked to prepaes and defaults of the reference mortgage loans, the securities are debentures, not mortgage-backed securities. Freddie's risk-transfer shelf is Structured Agency Credit Risk (STACR), while Fannie's is Connecticut Avenue Securities (CAS).

The first STACR and CAS deals both contained two mezzanine classes with LIBOR floaters of 1.35%

John Sim (1-212) 834-3124
Abhishek Mistry^{AC} (1-212) 834-4662

thickness (Exhibit 1). The reference pool is split into the two mezzanine bonds as well as one senior and one bottom residual tranche of 30bp thickness. Although we refer to the senior bond and residual as classes receiving principal and interest, these are not actual bonds that are sold to investors. Rather, they are used only for the purposes of calculating the cash flows to M1 and M2; they can be thought of as effectively retained by Freddie Mac or Fannie Mae.

The cash flows rely on prepayments and credit events on the reference pool. Instead of traditional defaults, which are thought of as REO liquidation, short sales or foreclosure sales, writedowns will be allocated based on credit events, defined roughly as 180 day delinquency or short sale/REO, whichever comes first. In practice, this means that once a loan hits 180 day delinquency, it will drop out of the sample and create an immediate loss.

Losses flow through from the bottom up, first hitting the residual 30bp, then the M2 and M1. By issuing these notes, Fannie Mae effectively transfers the credit risk of a reference pool of loans to investors who buy the M1 and M2 notes of the deal. Principal and interest are paid sequentially within the subordinates, although triggers can redirect principal from the subs to the senior (more on this later). The M1 tranche in these deals has generally been rated BBB by the ratings agencies, while the M2 tranche was unrated.

Rather than using proceeds from liquidation, and to simplify the process, Freddie and Fannie follow specified severity schedules based on cumulative credit events (Exhibit 2). In the Freddie deals, the first 100bp of credit events is assigned a 15% severity. The next 100bp has a 25% severity; after that, the severity is 40%. Fannie's schedule is similar but lower, at 10%, 20%, and 40%.

Exhibit 2: Severities are given by pre-set schedules and losses are incurred when a "credit event" is triggered

Cum Credit Events	STACR	CAS
< 1.0%	15%	10%
1.0 - 2.0%	25%	20%
> 2.0%	40%	40%

Source: J.P. Morgan, deal documents

As stated before, the bonds have a 10-year maturity, at which point all remaining principal is paid out to investors. Additionally, the bonds could be called early if the balance falls below 10%.

Interest is paid pro-rata to both bonds. Unlike most non-agency structures, principal is not locked out from mezz bonds entirely; principal should generally be paid pro-rata between the senior and the subs. However, within the subs, principal is paid sequentially, with M1 receiving principal first.

Additionally, there are triggers that allow principal to be redirected away from the subs. For both STACR and CAS, the senior credit enhancement must be at least 3%. As losses come in and erode the balance of subordinate bonds, the credit enhancement will fall below this threshold. At that point, unscheduled principal will be redirected to the senior tranche until the enhancement returns above 3%. Note that since principal is ordinarily paid pro-rata, the deals will not de-lever simply over time due to prepayments, a large difference from most non-agency structures.

Unlike the CAS deal, STACR has an additional trigger. The second trigger is hit if cum credit events rise rapidly, ahead of a predetermined schedule of 0.25% per year. In this second trigger, principal is again redirected to the

Exhibit 3: M1 sees significant extension in stress scenarios due to the triggers. The net credit event trigger is less salient if defaults are very high, but the credit enhancement trigger is still important then

STACR 2013-DN1 M1 and M2 yield and average life under various HPA scenarios and triggers

	M1						M2					
	Triggers		No Net CE Trigger		No Triggers		Triggers		No Net CE Trigger		No Triggers	
	Yield	AL	Yield	AL	Yield	AL	Yield	AL	Yield	AL	Yield	AL
Base HPA	3.6%	2.5	3.6%	2.5	3.6%	2.3	7.3%	8.9	7.3%	8.9	7.3%	8.5
Stress HPA	3.6%	3.4	3.6%	3.4	3.6%	2.8	7.1%	9.8	7.1%	9.8	7.1%	9.5
Repeat Financial Crisis	1.6%	5.2	1.6%	5.1	0.0%	2.6	-34.6%	-	-34.6%	-	-34.6%	-
10CPR / 0.5CDR	3.6%	7.1	3.6%	2.9	3.6%	2.1	4.3%	9.9	4.2%	9.7	3.9%	8.2
8CPR / 0.75CDR	3.6%	7.7	3.6%	6.1	3.6%	2.8	-6.8%	9.9	-6.8%	9.9	-7.2%	9.4

Source: J.P. Morgan

John Sim (1-212) 834-3124
Abhishek Mistry^{AC} (1-212) 834-4662

senior class. This trigger will not be hit if defaults are small, but can be hit in a rapidly deteriorating housing market. In extreme scenarios, these bonds could receive very little principal.

The chief effect of these triggers is to extend the average life of both subordinate bonds, particularly M1. We ran cash flows on the first STACR deal with and without the triggers under a few housing scenarios and some constant speeds. Under the base scenario, the cash flows with triggers are 0.2 years longer on M1 and 0.4 years longer on M2 (Exhibit 3). Both of these arise from delevering caused by the credit enhancement trigger. In more stressed scenarios, the relative impact of the two triggers varies. For example, running 10CPR and 0.5CDR causes the credit enhancement trigger to be hit infrequently – this trigger only adds 0.8 years to the M1 tranche. But the net C.E. trigger adds over 4 years. If defaults are very high, the distinction diminishes as both triggers will be hit until maturity. For example, in a repeat of the financial crisis, the credit enhancement trigger causes almost as much extension as having both triggers, about 2.5 years of extension. Running 8CPR and 0.75CDR results in 3.3 years of extension on M1 due to the enhancement trigger and another 1.6 years due to the net C.E. trigger.

The first STACR deal of 2014 added an additional twist to the structure, with an additional mezzanine bond. STACR 2014-DN1 M2 and M3 are similar to the M1 and M2 in previous deals, but in addition there is a single A-rated M1 tranche. This has allowed Freddie to increase deal size, reaching \$1bn of bonds sold in the last deal, compared to \$500mm in the first one.

Collateral characteristics

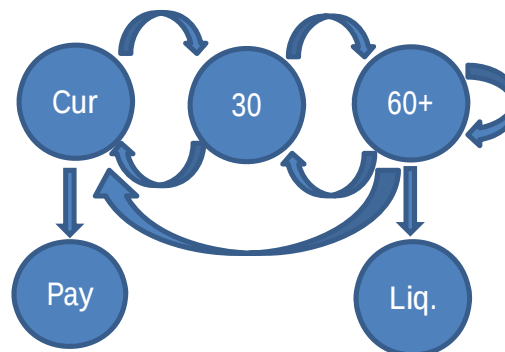
The underlying reference pools consist of 30-year fixed-rate mortgages wrapped by the GSEs, with particular credit constraints. Using the first STACR deal as an example, the underlying loans had the following characteristics, among others:

- 30 year fixed-rate with full documentation
- No mortgage insurance
- LTV between 60 and 80, inclusive of 80
- Combined LTV less than 95
- No HARP loans

Exhibit 4: With no need to track loans through foreclosure processing and liquidation, the transition model framework is greatly simplified

The arrow from and to sixty-plus state represents a likelihood of missing a payment while being at least 2 months past due and not charged off. Arrows represent models estimated.

"Pay" refers to voluntary payoff and "Liq" refers to charge off due to a credit event



Source: J.P. Morgan

Exhibit 5: Rolls are driven by loan attributes and pay history coupled with macroeconomic drivers, such as home prices and unemployment at an MSA level

As Of Dec-12 – FH Fixed 30 – Amortizing fully documented loans

Variables	Delinquency	Cure	Payoff	Credit Event
At Origination Attribute				
Original FICO	X	X	X	X
Original Combined LTV	X	X	X	X
Occupancy	X	X	X	X
Loan Purpose	X	X	X	X
Original Loan Size			X	
Dynamic Variables				
Loan Age	X	X	X	
Current Combined LTV	X	X	X	X
Months in state	X	X		X
Economic Refinancing Incentive			X	
Lagged Unemployment level	X			

Source: J.P. Morgan

- Not from a recourse state
- Never delinquent (loans were more than 9 months old)
- Balance of at least \$5,000 as of the month prior to issuance

The chief difference among the deals is origination period, but generally issuance has lagged origination by 3 quarters, so the loans are over nine months old by the time of issuance. This is also important because of the

^{AC} Indicates certifying analyst. See last page for analyst certification and important disclosures.

John Sim (1-212) 834-3124
Abhishek Mistry^{AC} (1-212) 834-4662

never delinquent requirement; these loans have 9 months of current pay history.

STACR and CAS deals use very similar collateral requirements. A summary of each deal's structure and collateral can be found in Exhibit 16.

Credit model

To evaluate credit risk in GSE risk-sharing deals, we use a transition model framework. Our model is very similar to our non-agency model but simplified to reflect the 180-day chargeoff and fit using the Freddie delinquency data¹. We looked at fitting a model with the Fannie data as well, but since both were similar we chose to use only one model to be able to do consistent comparisons across deals. The Freddie data we used to estimate the model comprises conforming 30-year fixed-rate amortizing loans with full documentation. As with the STACR/CAS deals, pay history is provided until the loan goes to short sale/REO or reaches 180 days delinquent, at which point the loan drops out of sample.

We capture the delinquency pipeline with eight rolls (Exhibits 4-5). The roll rate models are used in our analytics platform to evolve the pipeline while simulating the delinquency distributions. This generates cash-flows after adjusting for projected losses. The eight roll rates that we estimated can be broadly broken down into four categories based on fundamental borrower behavior. On any given month, the borrower can either payoff the loan voluntarily, miss a payment and go delinquent, make up a payment and cure by one or more payments, and finally get charged off due to an involuntary terminal event.

The model voluntary prepayments are driven largely by economic incentive², loan size, credit scores, and appraisals as reflected in the current combined LTV. A point rally for a loan that is 25 basis points in the money can result in speeds increasing by a factor of 2.5.

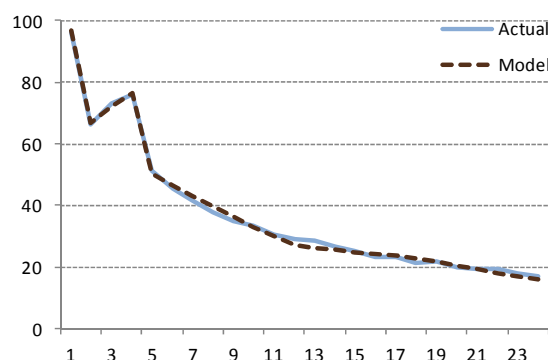
For modeling delinquency behavior, the three major drivers are FICO, current combined LTV, and pay history as represented by the number of consecutive months in state leading up to a credit event. Pay history is a key advantage of using our transition model

¹The full model write-up is available at <https://jpm.com/research/content/GPS-1215948-0>

² Incentive is measured as the difference between the note rate and prevailing mortgage rate lagged by 1 month

Exhibit 6: Borrowers reach the 180 day credit event quickly once entering 60+, but taper off thereafter

Probability of missing another payment while in 60+ by # of months in 60+



Source: J.P. Morgan, Freddie Mac

Exhibit 7: STACR 13-DN1 reference pool performance

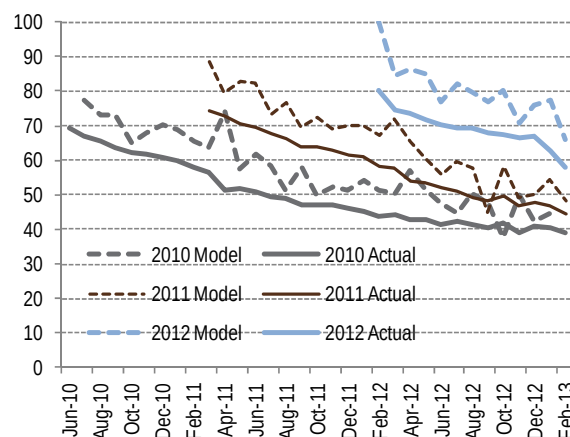
Roll rate (cur->30, 30->cur, etc) time series

Deal/Strats	Remit	CPR	Curr.-30	30-Curr.	30-60+	60+-C	60+-30	60+ & Making Pmt.	Credit Evt.
UPB(\$b): 21.7	Jul-13	0.00							
WAC: 3.84	Sep-13	6.0	0.08	84.1	4.0				
WALA: 17	Oct-13	4.4	0.12	82.0	6.1	7.3	7.9	21.5	0.0
FICO: 766	Nov-13	4.6	0.12	78.3	5.0	5.1	20.9	23.5	0.0
DTI: 32	Dec-13	4.3	0.10	72.1	6.0	12.7	20.8	21.9	0.0
CCLTV: 71	Jan-14	4.2	0.15	63.3	10.5	23.1	4.0	21.9	0.0
	Feb-14	3.9	0.14	72.0	5.8	15.5	14.7	18.4	0.0

Source: J.P. Morgan, Freddie Mac

Exhibit 8: Model performance on cure rates for recent vintages improves as they season

30d to current roll rate



Source: J.P. Morgan, Freddie Mac

framework. The probability of reaching later stages of delinquency drops off as the number of months in state

John Sim (1-212) 834-3124
Abhishek Mistry^{AC} (1-212) 834-4662

Exhibit 9: Our base case credit event projections are between 1.45% and 2.21% across the deals

Pricing as of 3/5			Fwd											Static							
	Bond		Yield	DM	AL	% Df1t	% WD	1st WD Date	3m CPR	1y CPR	Life CPR	Life CDR	Yield	DM	AL	3m CPR	1y CPR	Life CPR	Life CDR		
Base	STACR 2013-DN1	M1	2.84	174	2.3	1.50	0.0	..	12.3	11.3	7.7	0.25	1.55	136	1.8	12.3	12.1	11.1	0.2		
		M2	6.92	437	8.9	1.50	0.0	..	12.3	11.3	7.7	0.25	4.52	426	7.8	12.3	12.1	11.1	0.2		
	CAS 2013-C01	M1	2.51	145	2.3	1.45	0.0	..	12.4	11.4	7.8	0.24	1.52	133	1.9	12.5	12.2	11.2	0.2		
		M2	6.70	410	9.0	1.45	0.0	..	12.4	11.4	7.8	0.24	4.30	405	7.8	12.4	12.2	11.2	0.2		
	STACR 2013-DN2	M1	2.13	123	1.8	1.79	0.0	..	10.3	10.0	7.5	0.29	1.56	137	1.6	10.3	10.5	9.8	0.3		
		M2	6.53	382	8.6	1.79	1.7	Apr-22	10.3	10.0	7.5	0.29	4.36	410	7.8	10.3	10.5	9.8	0.3		
	CAS 2014-C01	M1	2.61	142	2.6	1.62	0.0	..	10.9	10.2	7.5	0.26	1.57	139	2.1	10.9	10.9	10.0	0.2		
		M2	6.79	411	9.3	1.62	0.0	..	10.9	10.2	7.5	0.26	4.37	411	8.5	10.9	10.9	10.0	0.2		
	STACR 2014-DN1	M1	1.82	85	2.4	2.21	0.0	..	10.4	9.7	7.0	0.35	1.01	84	1.9	10.4	10.3	9.7	0.3		
		M2	4.60	199	8.8	2.21	0.0	..	10.4	9.7	7.0	0.35	2.18	199	7.0	10.4	10.3	9.7	0.3		
		M3	6.03	331	9.7	2.21	10.3	Sep-19	10.4	9.7	7.0	0.35	3.91	367	9.8	10.4	10.3	9.7	0.3		
Depression	STACR 2013-DN1	M1	4.68	248	4.7	2.59	0.0	..	10.8	9.5	6.4	0.41	2.55	234	3.7	10.8	10.1	9.3	0.4		
		M2	4.98	245	8.8	2.59	21.2	Jun-18	10.8	9.5	6.4	0.41	3.37	314	9.0	10.8	10.1	9.3	0.4		
	CAS 2013-C01	M1	3.48	164	3.7	2.69	0.0	..	10.7	9.3	6.4	0.42	1.69	151	2.6	10.8	10.0	9.3	0.4		
		M2	5.11	249	9.2	2.69	18.6	Sep-19	10.7	9.3	6.4	0.42	3.08	286	8.9	10.7	10.0	9.3	0.4		
	STACR 2013-DN2	M1	3.62	127	4.7	3.31	0.0	..	9.1	8.3	6.2	0.51	1.61	142	4.6	9.1	8.7	8.0	0.5		
		M2	2.81	11	8.4	3.31	34.6	Jul-17	9.1	8.3	6.2	0.51	0.74	57	8.5	9.1	8.7	8.0	0.5		
	CAS 2014-C01	M1	3.65	149	4.6	3.05	0.0	..	9.3	8.3	6.1	0.47	1.65	146	3.4	9.3	8.8	8.2	0.4		
		M2	3.61	100	8.9	3.05	30.7	Sep-18	9.3	8.3	6.1	0.47	1.68	150	9.1	9.3	8.8	8.2	0.4		
	STACR 2014-DN1	M1	3.13	92	5.7	4.53	0.0	..	8.6	7.6	5.5	0.67	1.12	94	5.6	8.6	8.1	7.7	0.7		
		M2	4.79	201	10.0	4.53	0.0	..	8.6	7.6	5.5	0.67	2.24	204	10.0	8.6	8.1	7.7	0.7		
		M3	-2.68	-518	7.0	4.53	64.2	Oct-16	8.6	7.6	5.5	0.67	-3.89	-401	7.2	8.6	8.1	7.7	0.7		
Crisis	STACR 2013-DN1	M1	4.72	247	4.6	5.08	0.0	..	12.0	10.7	5.4	0.78	2.60	239	4.0	12.0	11.4	7.5	0.7		
		M2	-11.75	-1398	6.1	5.08	91.5	Jul-17	12.0	10.7	5.4	0.78	-9.39	-959	6.5	12.0	11.4	7.5	0.7		
	CAS 2013-C01	M1	2.92	69	4.7	5.70	6.1	Jan-23	12.0	10.7	5.2	0.86	1.87	168	4.1	12.1	11.4	7.4	0.8		
		M2	-22.67	-2512	5.8	5.70	100.0	Nov-17	12.0	10.7	5.2	0.86	-16.62	-1713	6.2	12.1	11.4	7.4	0.8		
	STACR 2013-DN2	M1	-4.52	-637	4.1	6.80	27.3	Jan-22	10.1	9.4	5.0	1.00	-1.77	-191	3.8	10.1	9.9	6.4	1.0		
		M2	-29.12	-3231	5.2	6.80	100.0	Feb-17	10.1	9.4	5.0	1.00	-34.91	-3733	5.4	10.1	9.9	6.4	1.0		
	CAS 2014-C01	M1	-4.08	-611	4.8	6.51	33.1	Feb-21	10.6	9.5	4.9	0.96	-3.77	-390	4.6	10.6	10.1	6.5	0.9		
		M2	-31.49	-3468	5.1	6.51	100.0	Jul-17	10.6	9.5	4.9	0.96	-37.91	-4083	5.2	10.6	10.1	6.5	0.9		
	STACR 2014-DN1	M1	3.18	92	5.5	8.85	0.0	..	10.0	8.9	4.5	1.28	1.11	94	4.8	10.0	9.5	6.1	1.3		
		M2	-5.84	-836	8.2	8.85	73.4	Jan-20	10.0	8.9	4.5	1.28	-6.07	-621	8.6	10.0	9.5	6.1	1.3		
		M3	-41.31	-4592	4.3	8.85	100.0	Oct-16	10.0	8.9	4.5	1.28	-47.70	-5261	4.4	10.0	9.5	6.1	1.3		

Source: J.P. Morgan

increases (Exhibit 6). The reasons for this can vary depending on the borrower-specific situation, especially around special cases like modifications or payment plans following a bankruptcy. In the case of a borrower that is neither modified, nor under distress, the fact that the borrower is making payments, albeit late, indicates a desire to remain in the home. Models using static loan-level attributes would not be able to pick up on this, but our model does display this behavior.

Cures can play an important role in any transition framework. The three major drivers of early cures are credit scores, CCLTV, and months in state. A borrower with a 780 FICO cures at a pace that is 17% faster than one with 690 FICO. A borrower with 70 CCLTV, or 30% equity, cures at a rate 35 % higher than a borrower that has no equity in the collateral. A loan that has remained 30-day delinquent, or rather has been making payments, while preserving the status of being one month behind, for a year, is 18 % less likely of curing than one that just fell behind a payment last month.

Cures are an area where the model may be biased by past experience and may not be fully capturing improvements in underwriting. We now have over six months of loan-level reporting data on the first STACR deal. Although we have seen a number of loans become delinquent, very few are making it to 60-day delinquency. After further examination, we found that roughly 70-80% of 30-day delinquent loans were curing (Exhibit 7). This compares to only 22% of prime fixed non-agencies 30-day delinquencies. Our GSE credit model projects roughly 50% cures on this deal, which is closer but clearly below the actual cures. Aligning the early stage rolls to actual STACR experience would lead to much lower projected credit events.

The discrepancies in borrower rolls highlight improvements in underwriting that we have seen over the past few years. The model fit cures better for more seasoned collateral (Exhibit 8). But there is an unquantifiable factor of improved underwriting, beyond

John Sim (1-212) 834-3124
Abhishek Mistry^{AC} (1-212) 834-4662

Exhibit 10: After controlling for FICO/LTV, the STACR deals consist of more cash-out in the larger slices

As of Jan-14 remits; CAS 14C01 (initial tape)

	FICO/LTV /CLTV	60-65	66-70	71-75	76-80 & CLTV<=80	76-80 & CLTV>80
CAS-14C01						
% UPB						
600-650	0.0%	0.1%	0.2%	0.4%	0.00%	
650-680	0.2%	0.4%	0.5%	1.2%	0.05%	
680-720	0.5%	1.6%	2.4%	5.0%	0.24%	
720-750	1.1%	2.1%	3.8%	7.2%	0.72%	
>=750	8.1%	11.0%	20.4%	30.2%	2.47%	
% Cash-out						
600-650	69.3	63.7	44.5	20.9		
650-680	29.5	26.2	38.9	21.7	0.0	
680-720	34.6	28.8	25.2	23.8	5.0	
720-750	23.0	22.8	20.1	15.5	20.2	
>=750	13.0	18.9	15.8	10.0	8.1	
ST-13DN02						
% UPB						
600-650	0.1%	0.1%	0.1%	0.2%	0.00%	
650-680	0.3%	0.5%	0.3%	0.7%	0.00%	
680-720	1.2%	2.1%	2.9%	5.0%	0.38%	
720-750	1.5%	2.3%	3.9%	7.1%	0.73%	
>=750	8.7%	12.0%	18.1%	29.2%	2.39%	
% Cash-out						
600-650	42.9	36.5	42.8	11.2	0.0	
650-680	4.2	48.5	33.9	7.3		
680-720	21.6	35.8	27.0	25.4	0.0	
720-750	27.3	33.2	24.0	23.6	0.0	
>=750	21.4	21.3	18.6	15.4	0.0	

Source: J.P. Morgan, Fannie Mae, Freddie Mac

what is measured by FICO and LTV, that the model is not capturing.

To be fair, we are not rushing to change the model. The GSEs are expected to widen the credit box on which these deals are done. As they do deals with high LTV or possibly HARP collateral, the model is likely to do a better job in its current form. But this is something to keep in mind when looking at model projections.

Default projections

Using the credit model, we project credit events and writedowns on the various bonds in three housing scenarios (Exhibit 9). The depression scenario has 15% worse home prices than base, while the crisis scenario repeats the downturn we saw during 2007-2010.

We project between 1.45% and 2.21% cumulative credit events across the deals in the base case. None of the senior mezz bonds take a writedown and the bottom mezz bond only takes a writedown in two of the deals.

Exhibit 11: Loans taken out for funding re-financings contribute to a larger share of defaults.

Base case cumulative defaults, as of Jan. 22, 2014

	% Cum. Def		% UPB	
	CAS- 14C01	STA- 13DN02	CAS- 14C01	STA- 13DN02
All	1.64%	1.75%		
Rate Refi				
>=720/>75	1.26%	1.45%	17.8%	18.0%
>=720/<=75	1.02%	1.12%	30.2%	29.9%
<720/>75	4.67%	4.00%	2.1%	2.5%
<720/<=75	3.77%	3.66%	2.9%	3.8%
	0.74%	0.83%		
Cashout				
>=720/>75	2.19%	2.14%	4.5%	6.2%
>=720/<=75	1.67%	1.59%	7.9%	9.9%
<720/>75	5.66%	5.85%	1.5%	1.3%
<720/<=75	4.95%	4.54%	1.8%	2.2%
	0.41%	0.47%		
Purchase				
>=720/>75	1.28%	1.40%	18.3%	15.3%
>=720/<=75	1.04%	1.16%	8.4%	6.8%
<720/>75	3.88%	4.07%	3.2%	2.6%
<720/<=75	3.43%	3.16%	1.3%	1.5%
	0.49%	0.45%		

Source: J.P. Morgan, Fannie Mae, Freddie Mac

This points to the very high credit of the collateral. There are so few losses that the 30bp residual is enough of a cushion. This is particularly true once you account for the better curing we have seen, as we discussed above. Better underwriting should lead to few writedowns on any of the deals so far in a base case.

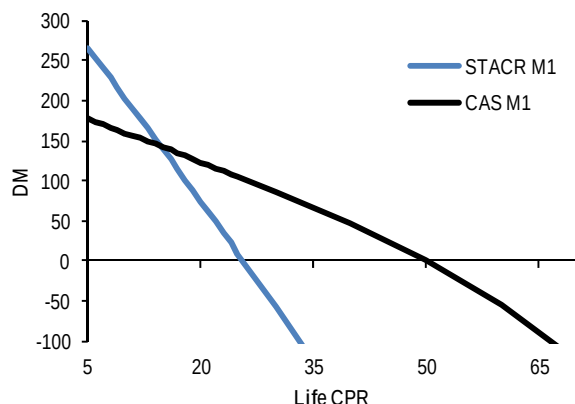
In downturn scenarios we see a hit to the bottom tranche of each of the deals, with M1 incurring writedowns in a repeat of the financial crisis.

The differences among deals are interesting, especially given that the collateral on the surface seems highly similar. For example, our model projects higher credit events on STACR 2013-DN2 compared to CAS 2014-C01. The culprit appears to be a higher concentration of refi loans, which contribute to a larger share of defaults (Exhibits 10-11).

John Sim (1-212) 834-3124
Abhishek Mistry^{AC} (1-212) 834-4662

Exhibit 12: STACR 2013-DN1 M1 has considerably more prepay convexity than CAS 2013-C01 M1

DM using JPM model defaults and given prepay speed. Pricing as of 10/23



Source: J.P. Morgan

Prepay analysis

A concern with buying GSE risk-sharing bonds is the prepay risk given that most are trading as premiums. This is particularly the case with the earlier deals, which were originated at a much higher spread and hence are very high premiums. The impact of prepaids varies quite a bit; for example, the first STACR deal is much more sensitive to prepaids than the first CAS deal (Exhibit 12).

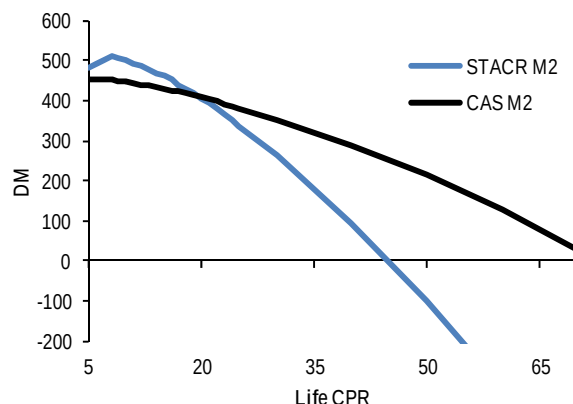
Lockout can counter some of the prepay risk. Recall that these bonds are sequential pays. The bottom tranche is locked out for a long time, mitigating some of the impact of fast speeds (Exhibit 13). While the STACR 2013-DN1 M1 spread was wiped out below 30 CPR, speeds need to be higher than 40CPR to wipe out the M2 spread.

Further insight comes from our pool-level prepayment model. In addition to looking at credit through the transition model framework, we are also able to run the collateral through a more traditional agency framework where prepayments are projected using refi incentive curves and pool-level attributes and a buyout model is used to estimate credit events. The advantage of this approach is that it allows us to run OAS and get a comparison of prepay sensitivity compared to agency TBAs, since both are evaluated using similar frameworks.

According to our model, the collateral pays similarly to TBA 3.5s (Exhibit 17). Given the WAC, which ranges from 3.60% to 3.84%, we think this makes sense. Speeds reach around 20CPR in a down 100bp scenario.

Exhibit 13: The M2 of both structures is less sensitive to prepaids due to lockout

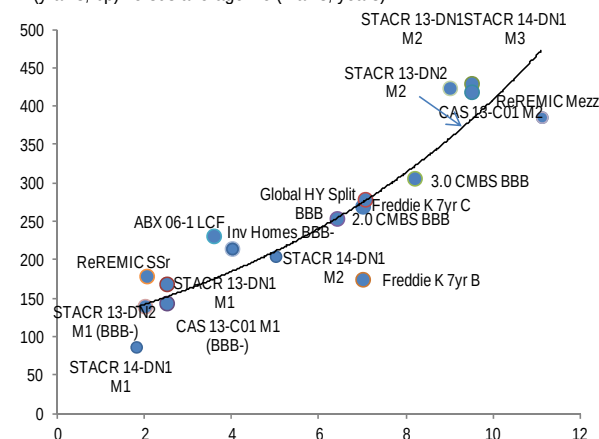
DM using JPM model defaults and given prepay speed. Pricing as of 10/23



Source: J.P. Morgan

Exhibit 14: Comparing GSE credit deals versus other assets

DM (y-axis, bp) versus average life (x-axis, years)



Source: J.P. Morgan

Relative value

From a relative value standpoint, we look at the GSE credit deals versus other forms of structured credit: legacy RMBS, CLOs, and CMBS (Exhibit 14). The M1 tranche of most deals has comparable average life and spread to Re-REMIC seniors and the Invitation Homes REO-Rental deal. The M2 tranches are more similar to CMBS 3.0 BBBs.

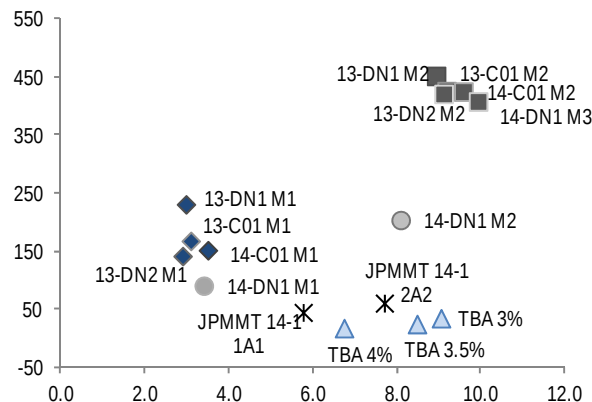
Looking at interest rate risk, it's also possible to calculate OAS with our pool-level model (Exhibit 15). This may be particularly useful for the M1 tranche of STACR 2014-DN1, which is single-A rated and less credit sensitive than the other bonds. It trades at a higher spread and shorter average life than 2.0 non-agencies or TBAs.

A key consideration with these comparisons is the liquidity differential among the various assets. GSE credit deals have seen 3-5 tick markets. CMBS and legacy RMBS are clearly less liquid, while agency TBAs are much more liquid. Non-agency 2.0 RMBS can vary based on shelf and dealer; some bonds may be as liquid as 8 ticks but others have much wider bid/asks.

Overall, the GSE credit deals offer an excellent way for investors to obtain mortgage credit exposure. Structures are highly standardized and transparent, and the size of each deal offers much larger exposure to BBB and below assets than many other sectors. We expect this area to see growth over the near future as the GSEs increase the size and scope of their deals.

Exhibit 15: OAS on the various tranches in comparison to TBAs and AAAs

OAS vs average life



Source: J.P. Morgan

John Sim (1-212) 834-3124
Abhishek Mistry^{AC} (1-212) 834-4662

Exhibit 16: A summary of STACR and CAS structural and collateral differences

	STACR 2013-DN1	STACR 2013-DN2	CAS 2013-C01	CAS 2014-C01	STACR 2014-DN1
Size (\$MM)	\$ 500	\$ 630	\$ 675	\$ 750	\$ 1,008
Collateral Reference Pool	Freddie Q3 2012	Freddie Q4 2012/Q1 2013	Fannie Q3 2012	Fannie Q4 2012	Freddie Q2 2013
Maturity	10 years	10 years	10 years	10 years	10 years
Classes/Thickness	M1/135bp, M2/135bp	M1/105bp, M2/165bp	M1/135bp, M2/135bp	M1/135bp, M2/135bp	M1/100bp, M2/150bp, M3/170bp
Ratings	M1/M2 NR	M1 BBB-sf, M2 NR	M1 BBB- Fitch, M2 NR	M1 BBB- Fitch / Baa2 Mdy M2 NR	M1 A-sf, M2 BBB(sf), M3 - NR KBRA
Initial Credit Enhancement (First Loss)	30bp	30bp	30bp	30bp	30bp
Clean-up call	10%	10%	10%	10%	10%
Initial Pricing@10CPR (approximate)	(priced July 26th, 2013)	(pricing November 5, 2013)	(priced October 16, 2013)	indicative as of Jan 9, 2014	(pricing February 12, 2014)
M1	340DM@10CPR	145DM@10CPR	200DM@10CPR	170DM@10CPR*	100DM@10CPR
M2	715DM@10CPR	425DM@10CPR	525DM@10CPR	430DM@10CPR*	220DM@10CPR
M3					450DM@10CPR
Initial Pricing Average Life					
M1	2.0	1.5	2.1	2.1	1.7
M2	7.7	7.1	7.9	8.1	4.7
M3					9.363
Current Mid Market Pricing	(as of Mar 03 2013)	(as of Mar 03 2013)	(as of Mar 03 2013)	(as of Mar 03 2013)	(as of Mar 03 2013)
M1	155	134	137	135	87
M2	424	417	405	410	191
M3					419
JPMorgan Model Average Life					
M1	2.3	1.8	2.2	2.5	2.4
M2	8.9	8.6	8.9	9.3	8.8
M3					9.7
JPMorgan Model Expectations(Base)					
Prepayments	7.8CPR	7.6CPR	7.9CPR	7.6CPR	7.1CPR
Cum. Credit Event	1.50%	1.78%	1.48%	1.61%	2.21%
Collateral Loss (Base)	0.27%	0.34%	0.20%	0.22%	0.48%
Writedowns M1/M2/(M3) (Base)	0.0/0.0	0.0/1.5	0.0/0.0	0.0/0.0	0.0/0.0/10.2
Severity (based on cum. Credit event)					
<1%	15%	15%	10%	10%	15%
1-2%	25%	25%	20%	20%	25%
2+%	40%	40%	40%	40%	40%
Credit Event	180 days, or early disposition	180 days, or early disposition	180 days, or early disposition	180 days, or early disposition	180 days, or early disposition
Triggers					
Min Credit Enhancement	greater than or = 3%	greater than or = 3%	greater than or = 3%	greater than or = 3%	greater than or = 3%
Cum. Net Credit Event Test	Cap starts at 25bp and increases to 2.5% by year 10, increasing 25bp / year	Cap starts at 25bp and increases to 2.5% by year 10, increasing 25bp / year	None	None	Cap starts at 25bp and increases to 2.5% by year 10, increasing 25bp / year
Collateral Constraints					
Qualifying collateral	30-year amortizing, Full doc, 1-4 Family, 1st Lien	30-year amortizing, Full doc, 1-4 Family, 1st Lien	25/30-year amortizing, Full doc, 1-4 Family, 1st Lien	25/30-year amortizing, Full doc, 1-4 Family, 1st Lien	30-year amortizing, Full doc, 1-4 Family, 1st Lien
OLTV	60-80%	60-80%	60-80%	60-80%	60-80%
CLTV	no cap	no cap	<= 97%	<= 97%	no cap
Collateral Detail					
\$ Balance	\$21.89 billion (as of Nov '13)	\$35.01 billion (as of Nov '13)	\$26.46 billion (as of Nov '13)	\$29.31 billion (as of Nov '13)	\$32.44 billion (as of Dec '13)
Avg Current Loan Size	231,800	241,700	236,200	239,700	236,000
WAC	3.84%	3.60%	3.85%	3.64%	3.72%
WALA	14	8	13	13	7
FICO	766	764	765	765	761
DTI	32.2%	32.1%	31.80%	31.5%	33.0%
OLTV	74.8%	74.3%	74.9%	74.5%	74.8%
MTM CLTV	67.0%	68.5%	66.5%	66.7%	68.5%
% Purchase	40%	27%	41%	31%	41%
% Cashout	17%	20%	15%	16%	19%
% FICO<690	4.1%	4.3%	4.9%	5.0%	5.8%
% FICO<690 & OLTV>75	2.1%	2.0%	2.6%	2.6%	2.9%
% First time home buyer	10.4%	7.3%	12.1%	9.2%	11.0%
% Single Family	71.1%	66.9%	63.4%	62.5%	65.3%
% Condo	4.8%	5.5%	9.3%	9.7%	6.8%
% Retail	46.1%	40.2%	48.5%	47.0%	43.5%
% Investor	5.3%	6.9%	8.4%	8.7%	9.0%

Source: J.P. Morgan, Fannie Mae, Freddie Mac

John Sim (1-212) 834-3124

Abhishek Mistry^{AC} (1-212) 834-4662**Exhibit 17: STACR/CAS convexity looks similar to TBA 3.5s**

Pool-level model runs, as of 3/5

Bond	Scenario	3M CPR	1Y CPR	Life CPR	Life CDR	Dur	Cvx	Bond	Scenario	3M CPR	1Y CPR	Life CPR	Life CDR	Dur	Cvx
STACR 2013-DN1 M1	-100	7.5	14.5	17.1	0.2	-1.6		CAS 2014-C01 M2	-100	5.5	9.7	13.5	0.3	0.1	
	-75	6.7	11.0	13.9	0.2	-1.4			-75	5.1	7.5	10.5	0.3	0.2	
	-50	6.1	8.7	10.9	0.2	-1.2			-50	4.8	6.5	8.6	0.3	0.0	
	-25	5.7	7.3	8.8	0.2	-1.0			-25	4.6	5.7	7.6	0.3	-0.1	
	0	5.5	6.4	7.8	0.2	-0.8	-0.8		0	4.5	5.2	6.9	0.3	-0.1	-0.2
	+100	5.0	4.8	5.7	0.2	-0.3			+100	4.1	4.1	5.3	0.2	0.0	
STACR 2013-DN1 M2	-100	7.5	14.5	17.1	0.2	-2.5		STACR 2014-DN1 M1	-100	3.9	8.4	14.3	0.3	0.1	
	-75	6.7	11.0	13.9	0.2	-1.3			-75	3.4	6.1	10.7	0.3	0.2	
	-50	6.1	8.7	10.9	0.2	-1.0			-50	3.2	4.8	8.5	0.3	0.1	
	-25	5.7	7.3	8.8	0.2	-0.6			-25	3.0	4.2	7.4	0.3	-0.1	
	0	5.5	6.4	7.8	0.2	-0.2	-1.1		0	2.9	3.8	6.6	0.3	0.0	-0.1
	+100	5.0	4.8	5.7	0.2	0.4			+100	2.7	3.1	5.1	0.3	0.0	0.0
CAS 2013-C01 M1	-100	7.6	14.1	17.0	0.3	-0.5		STACR 2014-DN1 M2	-100	3.9	8.4	14.3	0.3	-0.2	
	-75	6.7	10.7	13.4	0.3	-0.3			-75	3.4	6.1	10.7	0.3	0.1	
	-50	6.1	8.4	10.6	0.3	-0.3			-50	3.2	4.8	8.5	0.3	-0.1	
	-25	5.8	7.2	8.8	0.3	-0.4			-25	3.0	4.2	7.4	0.3	-0.2	
	0	5.5	6.3	7.7	0.3	-0.3	-0.3		0	2.9	3.8	6.6	0.3	-0.1	-0.2
	+100	4.9	4.7	5.7	0.3	-0.1			+100	2.7	3.1	5.1	0.3	0.0	
CAS 2013-C01 M2	-100	7.6	14.1	17.0	0.3	-0.9		STACR 2014-DN1 M3	-100	3.9	8.4	14.3	0.3	1.4	
	-75	6.7	10.7	13.4	0.3	-0.3			-75	3.4	6.1	10.7	0.3	1.4	
	-50	6.1	8.4	10.6	0.3	-0.4			-50	3.2	4.8	8.5	0.3	1.0	
	-25	5.8	7.2	8.8	0.3	-0.3			-25	3.0	4.2	7.4	0.3	0.7	
	0	5.5	6.3	7.7	0.3	-0.1	-0.5		0	2.9	3.8	6.6	0.3	0.6	0.4
	+100	4.9	4.7	5.7	0.3	0.2			+100	2.7	3.1	5.1	0.3	0.3	
STACR 2013-DN2 M1	-100	3.7	7.1	12.4	0.3	0.1		TBA FN 3%	-100	4.3	8.0	13.1	0.0	6.5	
	-75	3.5	5.6	9.7	0.3	0.2			-75	3.8	5.8	10.3	0.0	7.0	
	-50	3.3	4.9	8.1	0.2	0.1			-50	3.5	4.9	8.6	0.0	7.1	
	-25	3.3	4.4	7.2	0.2	-0.1			-25	3.2	4.3	7.7	0.0	7.3	
	0	3.2	4.1	6.5	0.2	-0.1	0.0		0	3.1	3.9	7.0	0.0	7.5	0.0
	+100	3.1	3.4	5.1	0.2	0.0			+100	2.9	3.2	5.7	0.0	7.7	
STACR 2013-DN2 M2	-100	3.7	7.1	12.4	0.3	0.1		TBA FN 3.5%	-100	5.5	12.0	17.2	0.0	5.0	
	-75	3.5	5.6	9.7	0.3	0.2			-75	4.3	8.1	14.4	0.0	5.8	
	-50	3.3	4.9	8.1	0.2	0.1			-50	3.4	5.7	11.5	0.0	6.2	
	-25	3.3	4.4	7.2	0.2	0.0			-25	3.0	4.5	9.4	0.0	6.6	
	0	3.2	4.1	6.5	0.2	0.0	-0.1		0	2.8	3.8	8.1	0.0	6.9	-0.6
	+100	3.1	3.4	5.1	0.2	0.1			+100	2.5	2.9	6.1	0.0	7.5	
CAS 2014-C01 M1	-100	5.5	9.7	13.5	0.3	0.0		TBA FN 4%	-100	10.4	25.8	24.3	0.0	2.7	
	-75	5.1	7.5	10.5	0.3	0.1			-75	8.4	20.1	20.6	0.0	3.7	
	-50	4.8	6.5	8.6	0.3	0.0			-50	6.2	14.7	17.7	0.0	4.4	
	-25	4.6	5.7	7.6	0.3	-0.1			-25	4.3	9.3	14.9	0.0	5.0	
	0	4.5	5.2	6.9	0.3	-0.1	-0.1		0	2.8	5.2	11.5	0.0	5.8	-2.1
	+100	4.1	4.1	5.3	0.2	0.0			+100	1.8	2.1	6.6	0.0	7.1	

Source: J.P. Morgan

Disclosures

Analyst Certification: The research analyst(s) denoted by an "AC" on the cover of this report certifies (or, where multiple research analysts are primarily responsible for this report, the research analyst denoted by an "AC" on the cover or within the document individually certifies, with respect to each security or issuer that the research analyst covers in this research) that: (1) all of the views expressed in this report accurately reflect his or her personal views about any and all of the subject securities or issuers; and (2) no part of any of the research analyst's compensation was, is, or will be directly or indirectly related to the specific recommendations or views expressed by the research analyst(s) in this report. For all Korea-based research analysts listed on the front cover, they also certify, as per KOFIA requirements, that their analysis was made in good faith and that the views reflect their own opinion, without undue influence or intervention.

Company-Specific Disclosures: Important disclosures, including price charts, are available for compendium reports and all J.P. Morgan-covered companies by visiting <https://jpm.com/research/disclosures>, calling 1-800-477-0406, or e-mailing research.disclosure.inquiries@jpmorgan.com with your request. J.P. Morgan's Strategy, Technical, and Quantitative Research teams may screen companies not covered by J.P. Morgan. For important disclosures for these companies, please call 1-800-477-0406 or e-mail research.disclosure.inquiries@jpmorgan.com.

Analysts' Compensation: The research analysts responsible for the preparation of this report receive compensation based upon various factors, including the quality and accuracy of research, client feedback, competitive factors, and overall firm revenues.

Other Disclosures

J.P. Morgan ("JPM") is the global brand name for J.P. Morgan Securities LLC ("JPMS") and its affiliates worldwide. J.P. Morgan Cazenove is a marketing name for the U.K. investment banking businesses and EMEA cash equities and equity research businesses of JPMorgan Chase & Co. and its subsidiaries.

Options related research: If the information contained herein regards options related research, such information is available only to persons who have received the proper option risk disclosure documents. For a copy of the Option Clearing Corporation's Characteristics and Risks of Standardized Options, please contact your J.P. Morgan Representative or visit the OCC's website at <http://www.optionsclearing.com/publications/risks/riskstoc.pdf>

Legal Entities Disclosures

U.S.: JPMS is a member of NYSE, FINRA, SIPC and the NFA. JPMorgan Chase Bank, N.A. is a member of FDIC. **U.K.:** JPMorgan Chase N.A., London Branch, is authorised by the Prudential Regulation Authority and is subject to regulation by the Financial Conduct Authority and to limited regulation by the Prudential Regulation Authority. Details about the extent of our regulation by the Prudential Regulation Authority are available from J.P. Morgan on request. J.P. Morgan Securities plc (JPMS plc) is a member of the London Stock Exchange and is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority. Registered in England & Wales No. 2711006. Registered Office 25 Bank Street, London, E14 5JP. **South Africa:** J.P. Morgan Equities South Africa Proprietary Limited is a member of the Johannesburg Securities Exchange and is regulated by the Financial Services Board. **Hong Kong:** J.P. Morgan Securities (Asia Pacific) Limited (CE number AAJ321) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission in Hong Kong. **Korea:** J.P. Morgan Securities (Far East) Ltd, Seoul Branch, is regulated by the Korea Financial Supervisory Service. **Australia:** J.P. Morgan Australia Limited (JPMAL) (ABN 52 002 888 011/AFS Licence No: 238188) is regulated by ASIC and J.P. Morgan Securities Australia Limited (JPMAL) (ABN 61 003 245 234/AFS Licence No: 238066) is regulated by ASIC and is a Market, Clearing and Settlement Participant of ASX Limited and CHI-X. **Taiwan:** J.P. Morgan Securities (Taiwan) Limited is a participant of the Taiwan Stock Exchange (company-type) and regulated by the Taiwan Securities and Futures Bureau. **India:** J.P. Morgan India Private Limited, having its registered office at J.P. Morgan Tower, Off. C.S.T. Road, Kalina, Santacruz East, Mumbai - 400098, is a member of the National Stock Exchange of India Limited (SEBI Registration Number - INB 230675231/INF 230675231/INE 230675231) and Bombay Stock Exchange Limited (SEBI Registration Number - INB 010675237/INF 010675237) and is regulated by Securities and Exchange Board of India. **Thailand:** JPMorgan Securities (Thailand) Limited is a member of the Stock Exchange of Thailand and is regulated by the Ministry of Finance and the Securities and Exchange Commission. **Indonesia:** PT J.P. Morgan Securities Indonesia is a member of the Indonesia Stock Exchange and is regulated by the BAPEPAM LK. **Philippines:** J.P. Morgan Securities Philippines Inc. is a Trading Participant of the Philippine Stock Exchange and a member of the Securities Clearing Corporation of the Philippines and the Securities Investor Protection Fund. It is regulated by the Securities and Exchange Commission. **Brazil:** Banco J.P. Morgan S.A. is regulated by the Comissão de Valores Mobiliários (CVM) and by the Central Bank of Brazil. **Mexico:** J.P. Morgan Casa de Bolsa, S.A. de C.V., J.P. Morgan Grupo Financiero is a member of the Mexican Stock Exchange and authorized to act as a broker dealer by the National Banking and Securities Exchange Commission. **Singapore:** This material is issued and distributed in Singapore by J.P. Morgan Securities Singapore Private Limited (JPMSS) [MIC (P) 049/04/2013 and Co. Reg. No.: 199405335R] which is a member of the Singapore Exchange Securities Trading Limited and is regulated by the Monetary Authority of Singapore (MAS) and/or JPMorgan Chase Bank, N.A., Singapore branch (JPMCB Singapore) which is regulated by the MAS. **Japan:** JPMorgan Securities Japan Co., Ltd. is regulated by the Financial Services Agency in Japan. **Malaysia:** This material is issued and distributed in Malaysia by JPMorgan Securities (Malaysia) Sdn Bhd (18146-X) which is a Participating Organization of Bursa Malaysia Berhad and a holder of Capital Markets Services License issued by the Securities Commission in Malaysia. **Pakistan:** J. P. Morgan Pakistan Broking (Pvt.) Ltd is a member of the Karachi Stock Exchange and regulated by the Securities and Exchange Commission of Pakistan. **Saudi Arabia:** J.P. Morgan Saudi Arabia Ltd. is authorized by the Capital Market Authority of the Kingdom of Saudi Arabia (CMA) to carry out dealing as an agent, arranging, advising and custody, with respect to securities business under licence number 35-07079 and its registered address is at 8th Floor, Al-Faisaliyah Tower, King Fahad Road, P.O. Box 51907, Riyadh 11553, Kingdom of Saudi Arabia. **Dubai:** JPMorgan Chase Bank, N.A., Dubai Branch is regulated by the Dubai Financial Services Authority (DFSA) and its registered address is Dubai International Financial Centre - Building 3, Level 7, PO Box 506551, Dubai, UAE.

Country and Region Specific Disclosures

U.K. and European Economic Area (EEA): Unless specified to the contrary, issued and approved for distribution in the U.K. and the EEA by JPMS plc. Investment research issued by JPMS plc has been prepared in accordance with JPMS plc's policies for managing conflicts of interest arising as a result of publication and distribution of investment research. Many European regulators require a firm to establish, implement and maintain such a policy. This report has been issued in the U.K. only to persons of a kind described in Article 19 (5), 38, 47 and 49 of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (all such persons being referred to as "relevant persons"). This document must not be acted on or relied on by persons who are not relevant persons. Any investment or investment activity to which this document relates is only available to relevant persons and will be

^{AC} Indicates certifying analyst. See last page for analyst certification and important disclosures.

engaged in only with relevant persons. In other EEA countries, the report has been issued to persons regarded as professional investors (or equivalent) in their home jurisdiction. **Australia:** This material is issued and distributed by JPMSAL in Australia to "wholesale clients" only. This material does not take into account the specific investment objectives, financial situation or particular needs of the recipient. The recipient of this material must not distribute it to any third party or outside Australia without the prior written consent of JPMSAL. For the purposes of this paragraph the term "wholesale client" has the meaning given in section 761G of the Corporations Act 2001. **Germany:** This material is distributed in Germany by J.P. Morgan Securities plc, Frankfurt Branch and J.P.Morgan Chase Bank, N.A., Frankfurt Branch which are regulated by the Bundesanstalt für Finanzdienstleistungsaufsicht. **Hong Kong:** The 1% ownership disclosure as of the previous month end satisfies the requirements under Paragraph 16.5(a) of the Hong Kong Code of Conduct for Persons Licensed by or Registered with the Securities and Futures Commission. (For research published within the first ten days of the month, the disclosure may be based on the month end data from two months prior.) J.P. Morgan Broking (Hong Kong) Limited is the liquidity provider/market maker for derivative warrants, callable bull bear contracts and stock options listed on the Stock Exchange of Hong Kong Limited. An updated list can be found on HKEx website: <http://www.hkex.com.hk>. **Japan:** There is a risk that a loss may occur due to a change in the price of the shares in the case of share trading, and that a loss may occur due to the exchange rate in the case of foreign share trading. In the case of share trading, JPMorgan Securities Japan Co., Ltd., will be receiving a brokerage fee and consumption tax (shouhizei) calculated by multiplying the executed price by the commission rate which was individually agreed between JPMorgan Securities Japan Co., Ltd., and the customer in advance. Financial Instruments Firms: JPMorgan Securities Japan Co., Ltd., Kanto Local Finance Bureau (kinsho) No. 82 Participating Association / Japan Securities Dealers Association, The Financial Futures Association of Japan, Type II Financial Instruments Firms Association and Japan Investment Advisers Association. **Korea:** This report may have been edited or contributed to from time to time by affiliates of J.P. Morgan Securities (Far East) Ltd, Seoul Branch. **Singapore:** JPMS and/or its affiliates may have a holding in any of the securities discussed in this report; for securities where the holding is 1% or greater, the specific holding is disclosed in the Important Disclosures section above. **India:** For private circulation only, not for sale. **Pakistan:** For private circulation only, not for sale. **New Zealand:** This material is issued and distributed by JPMSAL in New Zealand only to persons whose principal business is the investment of money or who, in the course of and for the purposes of their business, habitually invest money. JPMSAL does not issue or distribute this material to members of "the public" as determined in accordance with section 3 of the Securities Act 1978. The recipient of this material must not distribute it to any third party or outside New Zealand without the prior written consent of JPMSAL. **Canada:** The information contained herein is not, and under no circumstances is to be construed as, a prospectus, an advertisement, a public offering, an offer to sell securities described herein, or solicitation of an offer to buy securities described herein, in Canada or any province or territory thereof. Any offer or sale of the securities described herein in Canada will be made only under an exemption from the requirements to file a prospectus with the relevant Canadian securities regulators and only by a dealer properly registered under applicable securities laws or, alternatively, pursuant to an exemption from the dealer registration requirement in the relevant province or territory of Canada in which such offer or sale is made. The information contained herein is under no circumstances to be construed as investment advice in any province or territory of Canada and is not tailored to the needs of the recipient. To the extent that the information contained herein references securities of an issuer incorporated, formed or created under the laws of Canada or a province or territory of Canada, any trades in such securities must be conducted through a dealer registered in Canada. No securities commission or similar regulatory authority in Canada has reviewed or in any way passed judgment upon these materials, the information contained herein or the merits of the securities described herein, and any representation to the contrary is an offence. **Dubai:** This report has been issued to persons regarded as professional clients as defined under the DFSA rules. **Brazil:** Ombudsman J.P. Morgan: 0800-7700847 / ouvidoria.jp.morgan@jpmorgan.com.

General: Additional information is available upon request. Information has been obtained from sources believed to be reliable but JPMorgan Chase & Co. or its affiliates and/or subsidiaries (collectively J.P. Morgan) do not warrant its completeness or accuracy except with respect to any disclosures relative to JPMS and/or its affiliates and the analyst's involvement with the issuer that is the subject of the research. All pricing is as of the close of market for the securities discussed, unless otherwise stated. Opinions and estimates constitute our judgment as of the date of this material and are subject to change without notice. Past performance is not indicative of future results. This material is not intended as an offer or solicitation for the purchase or sale of any financial instrument. The opinions and recommendations herein do not take into account individual client circumstances, objectives, or needs and are not intended as recommendations of particular securities, financial instruments or strategies to particular clients. The recipient of this report must make its own independent decisions regarding any securities or financial instruments mentioned herein. JPMS distributes in the U.S. research published by non-U.S. affiliates and accepts responsibility for its contents. Periodic updates may be provided on companies/industries based on company specific developments or announcements, market conditions or any other publicly available information. Clients should contact analysts and execute transactions through a J.P. Morgan subsidiary or affiliate in their home jurisdiction unless governing law permits otherwise.

"Other Disclosures" last revised December 7, 2013.

Copyright 2014 JPMorgan Chase & Co. All rights reserved. This report or any portion hereof may not be reprinted, sold or redistributed without the written consent of J.P. Morgan.